

Capstone Project: E-Commerce Website Automation Framework (Chroma)

Project Title:

Automation Framework for Testing Croma E-Commerce Website

Project Description:

This project aims to create a comprehensive automation framework that tests key functionalities of the Croma e-commerce website. The focus is on validating search, filtering, sorting, and product pricing features. The framework will be developed using TestNG for test execution, Maven for project management, Extent Reports for reporting, and additional tools like Excel for data-driven testing, screenshot capture, and log file generation.

Domain Description:

The project is within the e-commerce domain, focusing on the online retail industry. The Croma website provides a platform for purchasing electronic goods, making it crucial to ensure the accuracy of its search, filter, and sorting functionalities. By automating these processes, the goal is to improve reliability, user experience, and operational efficiency.

Tools and Technologies:

- **TestNG:** Used for structuring and executing automated test cases.
- **Maven:** For project management, dependency management, and build automation.
- **Extent Reports:** To generate detailed and interactive test reports.
- **Excel Sheets:** To facilitate data-driven testing by storing input and expected output data.
- **Screenshots:** To capture images of the web pages at key test execution points for validation.
- **Log Files:** To record execution steps, outcomes, and any errors encountered during testing.

Application: go to <https://www.croma.com/>

Test Scenarios:

1. Search Functionality:

- **Objective:** Verify that the search functionality returns relevant results when the user searches for "Refrigerator."
- **Implementation:**
 - Use Excel to input "Refrigerator" as the search term.
 - Validate that the search results are relevant to refrigerators.

2. Brand Filtering:

- **Objective:** Ensure that the website's brand filter works as expected when selecting multiple brands (Samsung, LG, Whirlpool).
- **Implementation:**
 - Apply the brand filters for Samsung, LG, and Whirlpool.
 - Verify that only products from these brands are displayed in the results.

3. Sorting by Discount:

- **Objective:** Test the sorting functionality to ensure products are listed in descending order of discount.
- **Implementation:**
 - Apply the discount sort filter.
 - Verify that the products are sorted correctly by discount percentage.

4. Price Calculation:

- **Objective:** Calculate and print the average price of the top 10 discounted products.
- **Implementation:**
 - Extract the prices of the top 10 products after sorting by discount.
 - Compute the average price and display it in the test output or report.

5. Screenshot Capture:

- **Objective:** Capture screenshots of each significant action, such as after searching, filtering, and sorting.
- **Implementation:**
 - Take screenshots at each critical step to ensure visual verification.

6. Log Generation:

- **Objective:** Maintain a detailed log of each step in the test execution to facilitate debugging and tracking.
- **Implementation:**
 - Generate logs that record the start and end of each test, along with outcomes and errors.

7. Report Generation:

- **Objective:** Produce a comprehensive test report using Extent Reports.
- **Implementation:**
 - Include test results, logs, and screenshots in the report for detailed analysis.

Additional Scenarios:

1. Add to Cart Functionality:

- **Objective:** Validate that users can add products to the cart and that the cart reflects the correct products and quantities.
- **Implementation:**
 - Select a product, add it to the cart, and verify that it appears in the cart with the correct details.

2. Product Comparison:

- **Objective:** Test the product comparison feature by selecting multiple products and ensuring accurate comparison details.
 - **Implementation:**
 - Choose a few refrigerators, use the comparison feature, and validate that the product details are correctly compared.
- 3. Checkout Process:**
- **Objective:** Validate the entire checkout process, from selecting a product to completing the purchase.
 - **Implementation:**
 - Proceed through the checkout steps, verify the payment options, and ensure that the order confirmation is generated correctly.
- 4. User Login and Registration:**
- **Objective:** Ensure that the login and registration processes function correctly.
 - **Implementation:**
 - Test the registration of a new user, login with the registered credentials, and validate that the user is redirected to the correct page after login.

Execution Steps:

- 1. Set Up Framework:**
 - Initialize a Maven project.
 - Add dependencies for TestNG, Selenium WebDriver, Extent Reports, Apache POI (for Excel), and Log4j (for logging).
- 2. Develop Test Cases:**
 - Implement TestNG classes for each scenario.
 - Use WebDriver to interact with the Croma website.
- 3. Data-Driven Testing:**
 - Implement methods to read data from Excel sheets for inputs like search terms and expected results.
- 4. Enhance Reporting:**
 - Configure Extent Reports to generate interactive reports, including screenshots and logs.
- 5. Test Execution:**
 - Run the tests, analyze the results, and refine the tests as necessary.
- 6. Continuous Integration:**
 - Optionally, set up a CI/CD pipeline to run tests automatically on code changes.

This comprehensive project will help you understand how to build a scalable and maintainable automation framework for testing an e-commerce platform, covering a broad range of functionalities.